

Topic 9: AI for Research and Information

9.1 Perplexity AI -- AI-Powered Search

- Perplexity AI is a search engine that combines real-time web search with AI-synthesised answers and cited sources
- Unlike standard AI chatbots, Perplexity always searches the web and shows exactly which sources it used
- Free tier: limited daily Pro searches; Perplexity Pro (\$20/month): unlimited searches, GPT-4o and Claude backends
- Pro Search: performs deeper multi-step research, analysing papers and pages in more detail
- Spaces: create a focused research environment with specific curated sources and a custom AI assistant
- Try it: <https://www.perplexity.ai>

9.2 AI for Academic Research

- Consensus (consensus.app): AI search engine for peer-reviewed research -- ask yes/no research questions and see the evidence
- Elicit (elicit.com): finds relevant papers, extracts key data, and synthesises research findings across sources
- Semantic Scholar (semanticscholar.org): free AI-enhanced academic search with paper summarisation and citation graphs
- SciSpace (scispace.com): upload any paper and chat with it; searches and explains millions of academic papers
- Suggested workflow: Perplexity for broad overview -> Consensus for academic evidence -> NotebookLM to synthesise papers

9.3 Fact-Checking AI Output

- Assume any specific factual claim from an AI tool could be wrong -- always verify before using
- For statistics: find and cite the original source, not the AI's claim
- For academic citations: search for the paper directly -- AI frequently invents plausible-sounding but non-existent citations
- For recent events: AI models have a training cutoff; use Perplexity AI or ChatGPT with browsing for current events
- Useful habit: ask the AI 'How confident are you in this? What is the primary source?' -- this surfaces uncertainty

9.4 AI Browsing and Web Analysis

- ChatGPT with browsing (Plus): performs real-time Bing searches; most useful for current events and recent data
 - Claude with URLs: can fetch and analyse specific web pages you provide in your prompt
 - Gemini: strong real-time Google Search integration across most interfaces
 - Browser extensions: Sider, MaxAI -- add an AI chat sidebar to any webpage; ask questions about the page you are reading
 - Use case: paste a competitor's product page URL and ask for a structured analysis of their positioning and messaging
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